

Deep Learning–Based Detection of Population-Level Stress Signals in Social Media -

Milestone E: Final Project Report

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Code Repository: <https://github.com/uvter/AIT716-Stress-Detection>

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1. Introduction

Psychological stress is traditionally evaluated through methods such as clinical interaction, surveys, interviews, and self-reporting. These methods remain essential, particularly when individual diagnosis or treatment is involved, but they are not necessarily designed to identify changes in stress-related discourse occurring across large online populations. Social-media platforms provide a different kind of evidence. Users continuously discuss employment, finances, education, relationships, health, technology, social conflict, and other conditions that may produce frustration, uncertainty, or psychological pressure. Natural language processing offers a way to examine these discussions at a scale that would be difficult to achieve manually.

This project investigates whether a deep learning model can identify linguistic stress signals in social-media text and serve as a foundation for future population-level analysis. The objective is deliberately narrower than diagnosing mental-health conditions. A Reddit post classified as stressful does not establish that its author has a psychological disorder, nor does the proportion of stressful posts provide a direct measurement of the mental health of society. Instead, the model detects language that resembles human-annotated expressions of stress and evaluates whether these predictions could eventually be aggregated to identify changes in stress-related discourse.

The project uses Dreddit, a Reddit dataset developed by Turcan and McKeown (2019) for stress classification. The supervised portion used in this study contains 3,553 labeled text segments across multiple Reddit communities. Of these observations, 1,857 are labeled as stress and 1,696 as non-stress, resulting in a relatively balanced binary classification problem. The dataset is particularly useful because its examples are more contextual than short social-media messages and include discussions involving relationships, anxiety, trauma, financial concerns, and other sources of stress.

A fine-tuned DistilBERT transformer was selected as the primary model. Transformer architectures are appropriate because stress is not always communicated through obvious keywords. Meaning may depend on negation, relationships among words, indirect descriptions, contextual framing, and the interaction of multiple sentences. A TF-IDF logistic-regression classifier was implemented as a traditional baseline, while a majority-class model established a minimum performance reference.

Beyond model performance, the project addresses a larger analytical question. If automated stress classification is eventually applied to current social-media streams, the most useful signal may not be the raw number of stressful posts. Online discussions can become disproportionately visible because a relatively small group reacts intensely, because an issue spreads across many otherwise unrelated communities, or because repetition and viral amplification magnify a concentrated reaction. Future population-level analysis should therefore distinguish the **intensity, prevalence, persistence, and amplification** of stress-related discourse instead of reducing all online activity to one stress score.

The completed experiment evaluates the first stage of such a system: determining whether contextual deep learning can reliably distinguish stress-related from non-stress-related language while identifying the technical, interpretive, and ethical limits that must be addressed before broader deployment.

2. Literature Review

Recent research demonstrates a continuing shift from manually engineered lexical features toward contextual transformers, domain-adapted language models, explainable systems, large language models, and multimodal approaches for mental-health analysis of social-media text. At the same time, contemporary studies increasingly recognize that high classification accuracy alone does not resolve problems involving noisy labels, platform bias, interpretability, generalizability, or the difference between linguistic indicators and clinically valid conclusions.

Thushari et al. (2023) examined psychological well-being using interactions from Reddit communities associated with depression, anxiety, bipolar disorder, and suicide-related discussion. Their study compared contextual and non-contextual classification approaches and found that MentalBERT outperformed several traditional baselines. Importantly, the authors also incorporated LIME-based explainability rather than treating classification performance as the sole objective. This is relevant to the present project because it demonstrates two advantages of contextual modeling. First, language associated with mental well-being can depend on semantic relationships that are difficult to represent using simple lexical frequency. Second, applications involving sensitive

psychological information require mechanisms for understanding what linguistic evidence influences model behavior.

Zhang et al. (2023) moved beyond conventional binary detection by developing a sentiment-guided transformer with severity-aware contrastive learning for depression classification. Their approach combines semantic and sentiment information and uses supervised contrastive learning to distinguish between adjacent severity levels. The study highlights an important limitation of binary mental-health classification: two texts assigned to the same broad category may express substantially different emotional states. Although the current project retains Dreddit's binary stress labels, this work suggests a future direction in which stress could be modeled continuously or ordinally rather than as a simple stressed/non-stressed distinction.

Recent research has also focused on dataset scale and domain adaptation. Raihan et al. (2024) introduced MentalHelp, a semi-supervised Reddit dataset containing approximately 14 million instances across several mental-health tasks. The dataset was constructed using an ensemble involving FLAN-T5, Disor-BERT, and Mental-BERT. Their work addresses a persistent problem in mental-health NLP: manually labeling sensitive psychological text is expensive and time-consuming, which leaves many supervised datasets relatively small. The Dreddit subset used in the present project contains only 3,553 labeled examples, so large semi-supervised resources offer one possible method of expanding future training data. However, automatically generated labels also introduce a new risk because errors made by the labeling models may propagate into downstream systems.

Generalization across language and cultural context represents another active area of research. Raihan et al. (2024) evaluated traditional machine-learning, deep-learning, and transformer approaches for identifying stress in Tamil and Telugu code-mixed social-media text. Fine-tuned Tamil-BERT and Telugu-BERT models achieved macro F1 scores of approximately .71 and .72, respectively. Their findings support the effectiveness of transformer representations for stress identification while demonstrating that language-specific and code-mixed environments remain challenging. This is particularly relevant to a population-level system because an English-only Reddit classifier cannot be assumed to represent linguistically diverse populations.

Hengle et al. (2024) further illustrate the limitations of simplifying psychological language into isolated categories. Their ANGST benchmark addresses depression-anxiety comorbidity using multi-label classification rather than assuming that one post must correspond to one condition. The benchmark contains expert-annotated and silver-labeled social-media posts and was evaluated using models ranging from MentalBERT to GPT-4. Despite the use of advanced models, the authors found that no evaluated system exceeded an F1 score of .72 on the more difficult multiclass comorbidity task. This result is important because it shows that increasingly capable language models do not eliminate ambiguity when categories overlap conceptually. Stress is similarly intertwined with anxiety, anger, grief, financial uncertainty, frustration, and other emotional states. Consequently, a strong stress classifier should not be interpreted as possessing a complete psychological understanding of the author.

More recent work has begun exploring large language models and explicit explanatory reasoning. Lan et al. (2025) investigated LLM-based depression detection from social-media histories while focusing on the integration of medical knowledge and explanations. Their work identifies two continuing weaknesses in many deep-learning approaches: distinguishing clinically meaningful depression from temporary sadness and producing explanations that are useful to human experts. Although the present project intentionally avoids clinical diagnosis, the same distinction applies to population-level stress research. A model must eventually distinguish persistent or broadly distributed pressure from short-term emotional reactions if aggregate results are to be meaningful.

A particularly relevant development is the work of Soufleri and Ananiadou (2025), who evaluated a multimodal stress-detection system directly on Dreddit. Their approach generated visual representations from text and combined them with textual information through CLIP. They reported 94.90% accuracy using frozen CLIP features and 98.41% accuracy following full fine-tuning. Their ablation analysis also showed strong performance for text-only representations but improved results when modalities were combined. This demonstrates that Dreddit remains relevant to current stress-detection research and that the field is advancing beyond conventional text-only transformer classification.

However, the multimodal results should also be interpreted cautiously. Synthesized images are not observations supplied by Reddit users; they are model-generated representations of the text. This introduces an additional transformation layer that may encode biases or artifacts from the generative model. Soufleri and Ananiadou (2025) similarly identify dependence on a single dataset and uncertainty about generalization to

other platforms as limitations. Therefore, higher benchmark accuracy does not necessarily imply that a system is more appropriate for population-level deployment.

Across these studies, several trends emerge. Transformer models remain strong foundations for mental-health and stress classification, but research is increasingly moving toward domain adaptation, multimodal representations, explanation, larger weakly supervised datasets, multilingual modeling, and LLM-based reasoning. At the same time, recurring research gaps include limited cross-platform validation, difficulty distinguishing overlapping emotional states, outdated or nonrepresentative data, and uncertainty about how individual text classifications should be aggregated into socially meaningful indicators. The present project occupies a deliberately conservative position within this literature: it tests whether a lightweight transformer materially improves stress classification over traditional lexical methods while keeping broader population-level claims outside the scope of the current classifier.

3. Methodology

Dataset and Data Preparation

The study uses the labeled Dreddit train and test files created for supervised stress classification. The original training file contains 2,838 instances, while the held-out test file contains 715, producing 3,553 labeled observations in total (Turcan & McKeown, 2019).

Label 1 represents stress and label 0 represents non-stress.

An initial data audit examined missing text, label values, class distribution, duplicate records, and overlap between the supplied training and test sets. No missing text or invalid

class labels were identified. Eighteen exact duplicate text records were identified within the training data, and three text records appeared in both the original training and test files. Because identical text occurring in both development and evaluation data could inflate performance, duplicate training examples were removed and any training records matching the held-out test text were excluded before model development.

Text preprocessing was intentionally conservative. URLs were replaced with a common placeholder and repeated whitespace was collapsed. Punctuation, contractions, negation, pronouns, and emotionally relevant language were preserved. Aggressive stemming, lemmatization, and stop-word removal were avoided because these transformations can remove contextual information that a pretrained transformer may use.

After cleaning, the development data were divided into training and validation subsets using an 80/20 stratified split with random seed 42. The provided 715-record test file remained isolated until final evaluation.

Baseline Models

Two reference models were implemented.

The first was a majority-class DummyClassifier, which predicts the most frequent class and establishes a minimum performance threshold. Because the dataset is only mildly imbalanced, a system that cannot substantially outperform this baseline has little practical classification value.

The second baseline used TF-IDF features with logistic regression. The vectorizer incorporated unigrams and bigrams, used a minimum document frequency of 2, limited the vocabulary to 10,000 features, and applied sublinear term-frequency scaling. Logistic regression used a regularization parameter of $C = 2.0$ and a maximum of 2,000 iterations. This model provides a more meaningful comparison because it can learn discriminative stress-related words and phrases without learning contextual transformer representations.

DistilBERT Architecture

The primary system fine-tunes distilbert-base-uncased for two-class sequence classification. DistilBERT was developed as a distilled version of BERT that preserves much of BERT's language understanding while reducing model size and inference requirements (Sanh et al., 2019). This tradeoff was appropriate for the project because the available dataset is modest and training was conducted in Google Colab rather than on dedicated large-scale infrastructure.

The tokenizer converts each text into subword token IDs and attention masks. Inputs were truncated to a maximum of 256 tokens, with dynamic padding applied within each batch. The pretrained contextual representation is passed to a sequence-classification head that produces logits for the two output classes. Cross-entropy loss was used during fine-tuning. Training used the Hugging Face Trainer framework and AdamW optimization. The principal configuration was:

- Learning rate: 2×10^{-5}

- Training batch size: 16
- Evaluation batch size: 32
- Epochs: 3
- Weight decay: 0.01
- Warmup ratio: 0.10
- Maximum sequence length: 256 tokens
- Gradient clipping: 1.0
- Early-stopping patience: 2 evaluation cycles
- Model-selection criterion: validation F1
- Random seed: 42

The best checkpoint according to validation F1 was restored at the conclusion of training.

Dynamic padding and mixed precision when GPU support was available were used to control computational cost.

Tools and Frameworks

The prototype was implemented in Python and designed to run in Google Colab using a T4 GPU. Hugging Face Transformers provided the tokenizer, pretrained sequence-classification model, training framework, and early-stopping utilities. PyTorch supported model execution, while pandas and NumPy were used for data processing. Scikit-learn supported dataset splitting, TF-IDF vectorization, logistic regression, the majority baseline,

and evaluation metrics. Matplotlib was used to generate learning curves, the confusion matrix, and ROC visualization.

Evaluation Strategy

Stress-class F1 was selected as the principal metric because it balances precision and recall. Accuracy was also reported because the class distribution is relatively balanced, but accuracy alone does not distinguish between false-positive and false-negative behavior.

Precision measures the proportion of stress predictions that are actually labeled stress.

Recall measures the proportion of labeled stress examples successfully identified. Recall is especially relevant to a signal-detection application because a low-recall model could miss meaningful increases in stress-related discourse. However, maximizing recall without considering precision could cause anger, sarcasm, temporary outrage, or descriptions of difficult events to be interpreted incorrectly as stress. ROC-AUC was therefore included as an additional measure of ranking performance across classification thresholds.

The final evaluation also included a confusion matrix and qualitative error analysis. These outputs help reveal the kinds of mistakes that a single aggregate performance score cannot explain.

Figure 1. Recommended project workflow diagram

Dreaddit Data → Data Audit → Duplicate/Leakage Removal → Conservative Text Cleaning → Stratified Train/Validation Split → TF-IDF Baseline + DistilBERT Fine-Tuning → Validation

Model Selection → Held-Out Test Evaluation → Error Analysis → Future Population-Level Aggregation

4. Results

The DistilBERT model demonstrated useful learning across the three training epochs. Validation F1 increased to .8231 by epoch 3, while validation accuracy reached .8103. Validation precision for the stress class was .8084, recall was .8384, and ROC-AUC was .8938.

Table 1

Validation Performance Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Majority-class baseline	.5266	—	—	—	—
TF-IDF logistic regression	.7677	.7695	.7980	.7835	.8564
DistilBERT	.8103	.8084	.8384	.8231	.8938

On the untouched 715-record test set, DistilBERT maintained similar performance, suggesting that validation results generalized reasonably to unseen examples.

Table 2

Held-Out Test Performance for DistilBERT

Metric	Result
Accuracy	.8084
Stress precision	.7915
Stress recall	.8537
Stress F1	.8214
ROC-AUC	.8783

The test confusion matrix contained 263 true negatives, 83 false positives, 54 false negatives, and 315 true positives. Stress recall of .8537 was higher than stress precision of .7915. The model therefore missed fewer stress examples than it incorrectly flagged as stress, which is consistent with a signal-detection orientation but creates an important interpretive limitation.

On the common validation set, DistilBERT improved stress-class F1 by approximately 3.96 percentage points over TF-IDF logistic regression (.8231 compared with .7835). ROC-AUC similarly increased from .8564 to .8938. The improvement supports the hypothesis that contextual transformer representations capture useful information beyond unigram and bigram frequency.

The learning curves also revealed a less straightforward result. Validation F1 improved from approximately .8129 at epoch 2 to .8231 at epoch 3, but validation loss increased from .4294 to .4711. Thus, the third epoch improved discrete classification outcomes while slightly worsening the model's validation loss. This may indicate emerging overfitting or less reliable probability calibration even though thresholded classification improved.

Qualitative error analysis showed that mistakes were concentrated in several communities involving relationships, anxiety, PTSD, and survivors of abuse. These domains naturally contain negative or difficult subject matter even when individual passages are labeled non-stress. This helps explain some of the 83 false positives: describing a difficult circumstance is not always equivalent to expressing stress.

Figure 2. Validation accuracy and F1 by epoch.

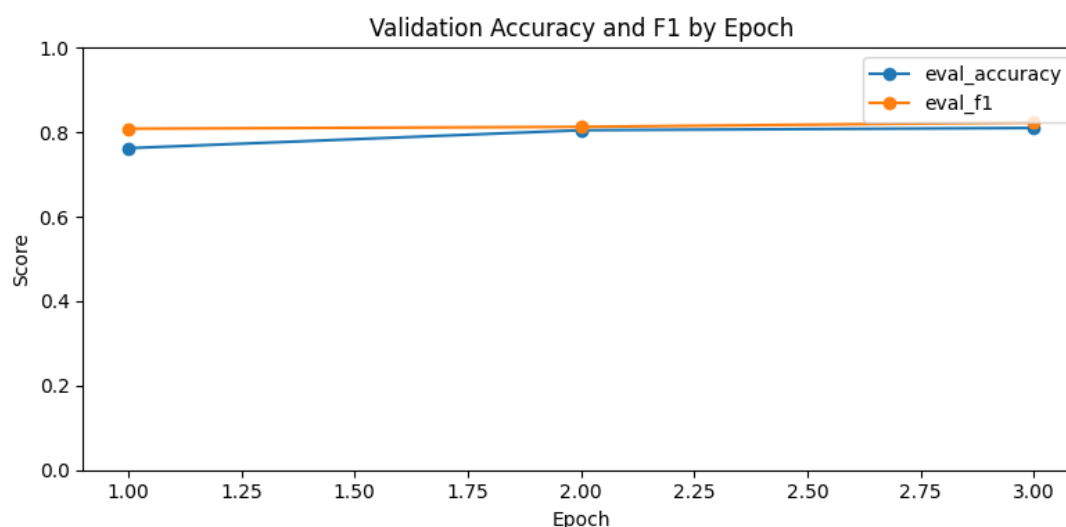
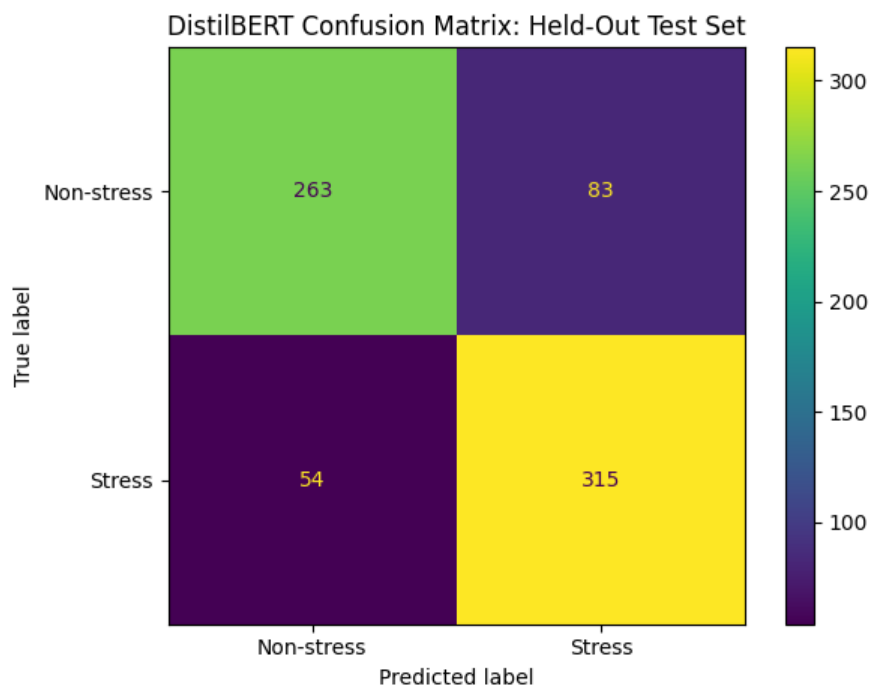


Figure 3. DistilBERT confusion matrix on the held-out test set.



5. Discussion

The primary strength of the project is that the transformer produced a measurable improvement over a meaningful traditional baseline while remaining computationally manageable. TF-IDF logistic regression already performed reasonably well, with validation F1 of .7835, showing that stress-related language contains useful lexical patterns.

DistilBERT raised F1 to .8231 on the same validation split and produced .8214 F1 on the held-out test set. This indicates that contextual information contributes additional value rather than the task being solvable only through obvious stress-related words.

A second strength is the attention given to data leakage. Discovering duplicate training observations and three train/test overlaps demonstrated that model evaluation depends on dataset quality as much as architecture selection. Had overlapping text remained in the

training data, the resulting evaluation could have overstated generalization. Removing these records reduced the available data slightly but produced a more credible experiment.

Several lessons emerged during development. First, a more advanced model does not eliminate ambiguity in the underlying labels. DistilBERT achieved relatively strong recall, but its 83 false positives show how easily difficult subject matter can be interpreted as direct stress. A post may discuss trauma, abuse, financial problems, or another person's distress without expressing the author's own stress in the way the dataset annotation expects. The model is therefore learning a useful but imperfect boundary between overlapping linguistic concepts.

Second, model-selection metrics can provide competing signals. Epoch 3 produced the strongest validation F1 while also showing increased validation loss. If the only objective were maximizing F1, epoch 3 would appear clearly preferable. If reliable confidence estimates were important, rising loss would justify additional calibration or comparison with the epoch-2 checkpoint. This illustrates why one metric should not define model quality.

The most important limitation is conceptual. This experiment classifies individual text segments, but the long-term research question concerns population-level stress signals. Those are not the same problem. Counting every positive prediction would allow a highly active or emotionally intense minority to dominate the apparent signal. Online visibility is not equivalent to population prevalence.

Future work should therefore distinguish at least four characteristics of stress-related discourse: **intensity, prevalence, persistence, and amplification**. Intensity represents how strongly stress-related language is expressed. Prevalence represents how widely the signal is distributed across distinct users or communities. Persistence captures whether an increase lasts briefly or continues over time. Amplification estimates whether a concentrated reaction is receiving disproportionate repetition or visibility.

This distinction is useful because a loud minority should not automatically be discarded as statistical noise. A strongly reacting subgroup may be identifying a legitimate source of anxiety or perceived threat before the concern becomes widespread. At the same time, minority voices can be intentionally or algorithmically amplified, allowing an issue to appear more representative than it actually is. The appropriate analytical response is therefore neither to accept amplified discourse as population sentiment nor to ignore it. The system should identify the emotional signal and then attempt to explain why that signal is amplified.

Perceived **loss of agency or control** may be one useful explanatory factor. Consumer arguments concerning physical versus digital ownership provide an illustrative example. Many consumers have accepted digital distribution without substantial concern, while others respond strongly to the possibility of losing permanent access, resale rights, preservation, offline availability, or independence from a platform provider. A superficial classifier might identify only anger or negativity. A richer analytical system could investigate whether the repeated underlying theme is perceived loss of control.

The same concept could generalize beyond gaming to subscription services, automation, employment, privacy, housing, healthcare, banking, or other domains in which people perceive that decisions or resources previously under their control have shifted toward institutions or automated systems. This does not imply that people discussing these issues have a psychological disorder. Instead, recurring language about helplessness, uncertainty, distrust, lack of control, or inability to influence outcomes may help explain why particular topics generate stress-related discourse.

This suggests a future system that analyzes both the emotional character of language and the structure of the discussion producing it. A high-intensity but low-prevalence event might indicate a concentrated community reaction. Moderate intensity that appears across many unrelated communities could provide stronger evidence of a broad social shift. High intensity combined with high prevalence and persistence would represent a substantially different phenomenon. Amplification analysis could then determine whether the apparent scale is organic, repetitive, coordinated, or driven by a limited number of highly active participants.

Such a framework would also help address concept drift. Dreddit predates many current economic, technological, and social changes. Specific vocabulary will evolve, but broader concepts such as uncertainty, perceived loss, control, security, and trust may transfer more effectively across time. Future research could therefore combine periodically updated transformer models with topic modeling or representation learning to examine not only whether stress language is increasing, but what themes are associated with that increase.

Other technical limitations remain. The current model uses a maximum sequence length of 256 tokens, which may truncate useful information in long posts. Hyperparameter testing was intentionally limited, and no systematic comparison of several learning rates, sequence lengths, classification thresholds, or transformer architectures was completed. Confidence calibration and complete subgroup evaluation were also outside the prototype's scope.

Potential applications should therefore remain aggregate and research-oriented. Public-health researchers, universities, nonprofit organizations, social scientists, and possibly organizations studying consumer behavior could use anonymized trends as one additional source of evidence. A stress-signal system would be most useful as an early analytical indicator that prompts further investigation rather than as an automated authority deciding why people are stressed or what should be done about them.

6. Ethical Reflection

The ethical risks of this project arise largely from the gap between what the classifier can detect and what people may assume its predictions mean. Dreddit labels linguistic stress signals rather than providing clinical diagnoses. A positive prediction should therefore never be interpreted as proof that an identifiable individual is psychologically distressed.

Privacy is a central concern. Social-media content may be publicly visible while still containing experiences that users did not expect to become inputs to automated psychological analysis. The present project reduces this risk by using an established dataset, excluding identifying fields from predictive features, and reporting results in

aggregate. A future deployment should follow the same principle and avoid creating persistent psychological profiles of individual users.

Bias also limits interpretation. Reddit is a self-selected platform population and does not represent society as a whole. Community membership, age distributions, linguistic culture, topic selection, and platform norms may all affect observed language. The dataset is also historical, increasing the possibility of concept drift. A classifier trained on earlier Reddit language may perform differently on newer slang, emerging concerns, or other platforms.

The observed false positives provide a concrete example of bias produced by context. Posts in anxiety, PTSD, relationship, or abuse-related communities may contain negative language because of the subject being discussed rather than because the passage expresses stress in the way the label defines it. If these errors were aggregated without adjustment, communities centered on difficult subjects could appear artificially more stressed.

Responsible use therefore requires transparency about uncertainty and error rates. The National Institute of Standards and Technology's AI Risk Management Framework emphasizes governing, mapping, measuring, and managing AI risks (NIST, 2023). In this project, that principle translates into clearly defining the model's purpose, measuring false-positive and false-negative behavior, monitoring changes in data, documenting limitations, and retaining human responsibility for interpretation.

The ACM Code of Ethics similarly emphasizes avoiding harm, fairness, transparency, and respect for privacy (Association for Computing Machinery [ACM], 2018). These principles

are especially relevant if social-media analytics are used by institutions with power over individuals. Employers, insurers, schools, governments, or law-enforcement organizations should not use this classifier to infer an individual's psychological condition or make high-stakes decisions.

Misuse can also occur at the population level. Amplified online activity can make a small number of voices appear representative of a much larger group. Conversely, dismissing a highly emotional minority as irrelevant could conceal an emerging issue. Responsible aggregation should therefore expose intensity and prevalence separately and provide evidence concerning persistence and amplification rather than generating a single authoritative societal stress score.

Future safeguards should include cross-platform validation, contemporary data collection, subgroup analysis where ethically possible, confidence calibration, drift monitoring, periodic retraining, and human review. Models should also be tested against manipulated or repetitive content so that coordinated amplification is less likely to distort aggregate interpretation.

There are situations in which not deploying the model would be the responsible decision. Individual diagnosis, automated crisis intervention, employment screening, insurance decisions, law-enforcement profiling, or surveillance exceed the evidence produced by this study. The model is best understood as an experimental linguistic sensor whose output can contribute to broader research when combined with other evidence and appropriate governance.

7. Conclusion

This project demonstrated that a lightweight pretrained transformer can effectively classify stress-related language in Reddit posts and outperform simpler lexical reference models. DistilBERT achieved a held-out stress-class F1 score of .8214 and ROC-AUC of .8783 while maintaining higher recall than precision. The result supports the usefulness of contextual language representations for identifying indirect and varied expressions of stress.

At the same time, the experiment demonstrates why accurate text classification is only the first step toward population-level social analysis. Reddit is not representative of society, Dreddit is historical, difficult subjects can generate false positives, and intense online activity may originate from a relatively small number of participants. Future systems should therefore treat stress as a multidimensional aggregate signal characterized by intensity, prevalence, persistence, and amplification.

A particularly promising research direction is examining themes associated with perceived agency and control. Strong reactions to changes in ownership, institutional trust, economic security, technology, or autonomy may reveal why certain topics produce concentrated stress-related discourse. The goal would not be to decide whether those concerns are objectively justified or to diagnose the individuals expressing them. Instead, AI could help identify where emotional pressure is emerging, how broadly it is distributed, how long it persists, and whether amplification is proportionate to the underlying population.

The completed prototype provides a technically functional foundation for that larger research direction while also demonstrating the interpretive and ethical safeguards required before social-media stress signals can responsibly support conclusions about communities or society.

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